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Cauliflower Leaf Disease Detection Using Computerized Techniques

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***Abstract*—Crop cultivation is an important role in the agriculture industry. Presently, food loss is primarily caused by sick crops, which affects growth rate and increase. High yield depends a lot on its growth. However, now crop leaf disease has become a significant problem in agriculture as a result of which the quality and growth rate of yield in agriculture is declining day by day. Through this paper, we have tried to diagnose the leaf disease of a particular crop (cauliflower). In this study, we collected data from each cauliflower leaf and created a dataset, and we divided our study was split into two sections: leaf disease detection and machine learning techniques.To detect leaf disease, we have used Matlab and various algorithms of classic machine learning such as Nav Bayes (NB), Decision Tree (TT), Random Forest (RF), Support Vector Machine (SVM), Secular Minimal Optimization (SMO). For each, we determined the accuracy, retraction, F-measurement, accuracy, and ROC values classification. This paper describes a simple but effective technique for reviewing leaf disease detection evidence and applying it to image processing and computer vision.**

***Index Terms*—Cauliflower, Leaf Disease, computer vision, SVM, k-Means**

I. INTRODUCTION

Bangladesh is known for its agriculture. Most of the people engaged in agriculture are 59.64 percent of the total population. The agriculture of a country plays an important role in its economic industry [1]. Compared to other countries in the world, Bangladesh is no longer behind. Bangladesh is ranked third in the world in terms of vegetable production [2]. Developed countries such as Japan, Norway, England, Sweden, the United States, and others have utilised advanced technology in agriculture, and it is a big shame that new technologies have not received much attention in Bangladesh's agricultural sector. Extensive fungal and bacterial diseases damage most plants. Farmers in Bangladesh do not have much knowledge about any crop disease so they are able to produce less crops. Call centers are available 24 * 7 for agricultural information and advice but sometimes they do not provide services and communication fails. Farmers are unable to accurately explain the disease in the affected areas due to a lack of analysis. Although our neighboring India makes extensive use of modern image processing features technology in order that they can convince their farmers about this. Foods from Cauliflower, Cucumber, Eggplant, cabbage, Lettuce, etc. [3]. Nowadays, the technology is widely used for predicting plant diseases and now Bangladesh has started using computerized methods. Which can assist farmers in this investigation with disease detection and with it a robotized structure has been created which can undoubtedly help farmers with diagnosis by taking pictures with camera. The concept of image processing with computer vision technologies helps us to:
i) Identification of infected leaves and stems
ii) Measure the affected area
iii) Find the shape of the infected area
iv) Determine the color of the infected area
v) and also affects the size and shape of the leaves.

The user must select a disease area on a leaf and then provide the cropped image for processing [4]. The goal of this paper is to investigate how to predict plant diseases at an early stage using the k-mean clustering algorithm then to detect the expected disease name, a multi SVM classifier was used to classify the data. We concentrate on diseases cauliflower leaf spots in particular. It could be useful for diagnosing Farmers' diseases [5]. It explains how to utilize image processing and data mining to analyze crop diseases and attributes. Furthermore, the contaminated area and affected percentage are calculated.

II. REVIEW OF EXISTING WORKS

Some researchers limit their studies to defective identification only, while others cover both identification and deficiency characteristics. However, not much effort has been made to detect plant diseases using computer vision with a specific focus on cauliflower leaf disease.
Nanehakaran etc al. [1] Leaf disease detection based on computer approach for recognition. Introduces a systematic method to verify the condition of tree leaves. The method is divided into two parts: image segmentation and image classification. For image detection, segmented images are used as the input of a synthetic neural network. And their accuracy stands at 85.59% and the error rate stands at 15.51%. Chouhan et al. [2] they proposed a research paper on leaf disease segmentation and classification using computer vision.. They used neural networks and super pixel clustering for segmentation. They extracted different algorithms are

used to create color, texture, and shape properties. And seven machine learning methods were used for classification purposes. Their accuracy is 98.57% and error rate stands at 1.43%.

M. U. Mojumdar et al. [3] SVM was used to present a computer vision method for wheat leaf disease pattern recognition. Using CV and image recognition, this paper presents a simple and qualitatively efficient technique for recognizing and evaluating evidence of leaf sickness.

Sambasivam et al. [4] SMOTE (Technique of Synthetic Minority Over-Sampling) and strategies for minimizing with focal failure deeply integrated networks of neurons built from the ground up are discussed in this paper. Their accuracy is 93% and error rate stands at 7%.

Sujatha et al. [5] the topic of this paper is performance of a deep understanding versus machine learning inside the detection of leaf diseases. To Classify citrus plant Disease they Used SVM, RF, SGD , and DL

Chethan et al. [6] the plant has implemented mobile applications for the Image processing and neural networks were used to classify plant leaf diseases. Here the clustering image was used to divide into clusters, and the characteristics of the diseased part were extracted using a gray layer co-presence matrix. Later, SVM and CNN were accustomed.

Jogekar et al. [7] this paper focuses on a study of in-depth learning methods for detecting and diagnosing plant leaf disease. A comparison of the output of deep learning methods for detecting and diagnosing different diseases in plant leaves using various plant leaf images is discussed here.

Gargade et al. [8] this research paper discusses Custard Apple Machine Learning for Apple Leaf Diseases Detection.which gives 99.5% accuracy and error rate 0.5%

Rodríguez et al. [9] They proposed a system, which is used to cherry bean detection on coffee trees. In Their work, 59% Accuracy Found for bourbon coffee plants with 0.594 accuracy;67% Accuracy Found for Cherry beans.

Tiwari et al. [10] they used classification as a tool in their survey work Potato Leaf Diseases Detection. Accuracy obtaining 97.8% and error rate 2.2%.

Hritik Arora et al. [11] Overall, this paper provides a great way to diagnose grain diseases using automated networks on a huge global scale, the process of training resonate models in open image datasets. The proposed ResNet34 model accomplished a 99.4% accuracy on a test set and error rate 0.01%.

In the light of the above research, we can conclude that no research has been done to give deep recognition to the diagnosis of cauliflower leaves. Our research is a clear investigation into the current state of work and adequate work.

III. Research Method

This framework describes what has been used or what has been done to diagnose cauliflower leaf disease using computer vision [3]. An analysis of this idea has shown that anyone, for example, who has no idea about this cauliflower leaf disease, if they read this page or topic properly, will understand that the proposed framework on cauliflower leaf disease has now been introduced. Here is a detailed description of it. In the beginning, we do the work in a few steps. The first step is image processing and labeling. In this step, photos were pre-processed to be used as statistics to classify computer perspectives to achieve high-quality extraction. The area around the diseased part of the cauliflower vegetable is then manually illustrated. We have confirmed that data has been requested for learning all the contents of the image. We resized figure 256 * 256 to limit the training time. We have identified all the pictures as great diseases. Duplicate and unwanted images have also been removed from the data-set. To learn this way we must verify the datasets. The second step is the image segmentation process. In this step, Image partitioning is a usually used technique for digital picture processing and analysis that divides an image into more than one component or area, frequently based on the pixel homes of the image. image splitting may additionally contain clustering regions of pixels to split the nozzle from the history or to fit coloration or length. The third step is Future Extract: The values we find in the data set are through 13(Contrast, Correlation, Variance, Root Mean Square(RMS) etc.) future extracts. In This Table we shows that, feature extraction formula for diseases classifications.

Feature Extraction Formula

01. Contrast

$$\sum_{i=1}^{N}\sum_{j=i}^{N}(i-j)^{2}\,p(i,j)$$

02. Correlation

$$\frac{\sum_{i=1}^{N}\sum_{j=1}^{N}(i-\bar{x})(j-\bar{y})\,p(i,j)}{\sigma x\sigma y}$$

03. Energy

$$\sum_{i=1}^{N}\sum_{j=i}^{N}p(i,j)^{2}$$

04. Entropy

$$\sum_{i=1}^{N}\sum_{j=i}^{N}p(i,j)\lg p(i,j)$$

The test for the latest improved performance is a very important part of this study. We have already provided data set data training with the help of computer vision information for this purpose. We get the result of our total work through testing.

Thus, all categories of cauliflower leaf disease have been independently isolated. Here we have used various algorithms of computer vision and here using Confusion Matrix [12] we get the values of accuracy, precision, clear evidence of master

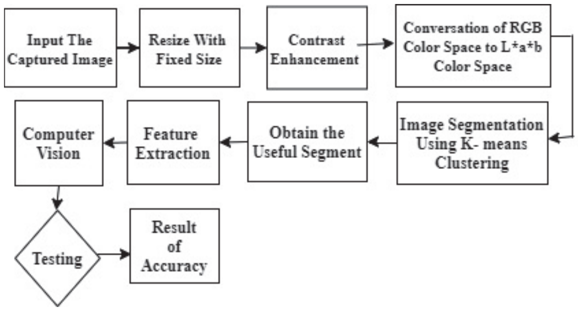

Fig. 1: Workflow diagram for the prediction model of a Cauliflower Leaf Disease Detection Using Computer Vision

structure, effectiveness, FPR and FNR.
Accuracy(ACC) =

$$\frac{TP+TN}{TP+FN+FP+TN} = \frac{TP+TN}{P+N}$$

Specificity(TNR) =

$$\frac{TN}{FP+TN} = \frac{TN}{N}$$

Precision(PPV) =

$$\frac{TP}{TP+FP} = 1 - FDR$$

Sensitivity =

$$\frac{TP}{P} = \frac{TP}{TP+FN} = 1 - FNR$$

FNR =

$$\frac{FN}{FN+TP} = 1 - TPR$$

FPR =

$$\frac{FP}{TN+FP} = 1 - TPR$$

IV. Collection and Analysis of Data

Due to the investigation, different types with photographs were collected with the help of the main level in collecting pictures of farmers from different farms, and pictures of diseases were selected in 3 categories(Black Rot,Yellow Virus,Length insects) where different diseases were going on. Different sections. All these class images are also kept in the class which examines the train data and other data. All images are taken from Cauliflower Land to mobile. Try to find good quality images when collecting images. Not all images are distorted, some images are healthy.

With K-means Clustering We portioned our Image data-set.To classify the diseases,We Used In our Work 13 GLCM [13] Features like as Variation,Kurtosis,Homogeneity etc.With help of GLCM, we Foud out features of the Image Data-set. After That,Diseases (Black Rot,Yellow Virus,Length insects) are Classified By SVM(Support Vector Machine).There Multiple Classifier we used our proposed work ,Such j48,random forest,random tree zeroR,naive bayes.With those Classifier We Foud Out Accuracy , Precision, Re-call etc.

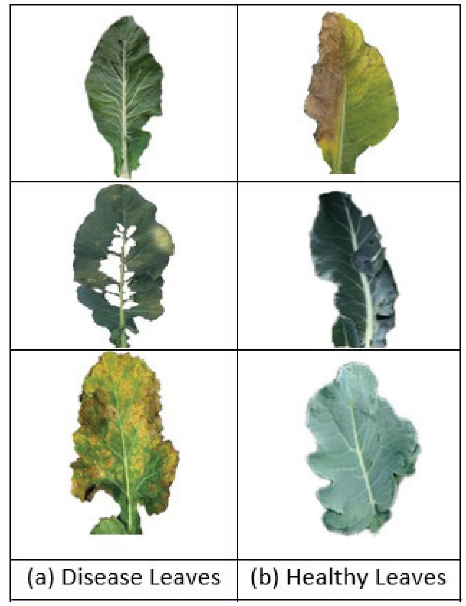

Fig. 2: Cauliflower Healthy & Unhealthy leaves

V. Result and Discussion

We've done a lot of work in this project worked With Black Rot,Yellow Virus,Length insects of Cauliflower.Several 866 data-sets used in our proposed system.We found 93% Accuracy with the help of Classifier.The table below shows the confusion matrix of leaf disease detection. beginfigure[htbp]

TABLE I: Confusion Matrix for Detecting Disease in Cauliflower Leaves

Cauliflower leaf Disease Detection			
Cauliflower leaf Disease		Predicted Class	
		Healthy	Infected
Actual Class	Healthy	631	8
	Infected	4	215

The figure below shows the bar chart of our work.

This detailed accuracy is done by LMT Classifier. Precision rate of Blackrot disease is 97%, Yellow virus disease is 96%, Leather insects disease is 95% and disease free 99%.

Detailed Accuracy By Individual Class							
TP Rate	**FP Rate**	**Precision**	**Recall**	**F-Measure**	**MCC**	**ROC Area**	**Class**
96%	8%	97%	96%	97%	96%	99%	Black rod
98%	12%	96%	98%	97%	96%	99%	Yellow Virus
95%	10%	96%	95%	96%	94%	99%	Leather Insects
100%	0%	1%	1%	1%	0%	1%	Disease Free

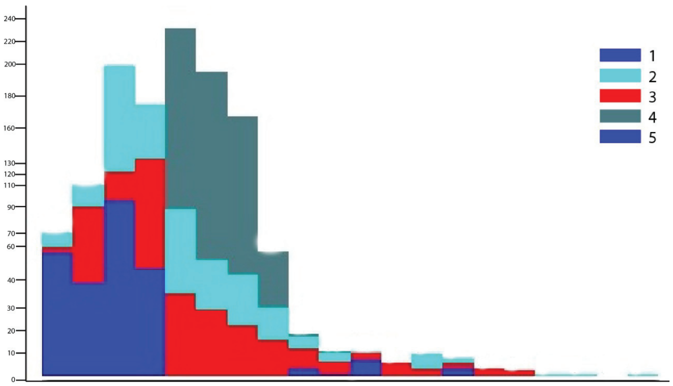

Fig. 3: Bar Chart in Cauliflower Leaves

The Sensitivity Ratio is a measure of how sensitive an individual is 95.43 %, implying that the ratio of perceiving the name of the disease Leaves away from the provided information is extremely high, resulting in an FNR of 4.56%.

Matrix	Outcome
Accuracy/Exactness	93.64%
Precision	96.31%
Specificity	96.56%
Sensitivity	95.43%
FNR	4.53%
FPR	3.47%

TABLE II: The Matrix Performance Results

Furthermore, the standard Specificity ratio is 96.56%, implying that the disease recognition ratio is sufficiently excellent.We also apply

Classifier	Accuracy
Random Tree	93.649%
Random Forest	90.569%
J48	91.1109%
Functions Logistic	75.1732%

TABLE III: Multiple Classifier Accuracy

In Table V we see the comparative results of the four classifications. The highest accuracy gives Random Tree 93.649% and lowest accuracy gives Functions Logistic 75.1732%.

VI. Conclusions

Our study has begun with a focus on the identification of cauliflower leaf diseases. To do this, we recruited a computer vision expert and used an administrative learning process to finalize the identification. We've finished 93 percent of the system's correctness, which is not only satisfactory but also commits us to more research in the future. For disease detection, we employed an SVM classifier.We employed a variety of algorithms to diagnose diseases. Plant diseases can be recognized early and pest management tools employed while minimising the risk to humans and the environment using this method.The following are the grounds for improper classification: The symptoms of diseased plant leaves vary (little brown to black dots at first, then dry leaf, black, or part leaf removal later), and feature identification vectors must be improved.To increase the rate of illness detection at various phases, training samples can be supplemented with ideal shape and color features, which can then be used as disease detection input conditions.

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